Besides the title and the abstract at the beginning, a paper can be written in many ways. However, before you are experienced, you may want to follow such a structure:

1. Introduction
2. Literature review
3. Research methodology or approach
4. Findings
5. Conclusion

More info: https://www.emeraldgrouppublishing.com/services/authors/author-how-guides/structure-your-journal-submission

Abstract

An abstract could be written in many ways, but it could be structured in this way if applicable:

- Purpose
- Design/methodology/approach
- Originality
- Findings
- Research/practical implications

More info: https://www.emeraldgrouppublishing.com/services/authors/author-how-guides/write-article-abstract

Introduction

Introduction should include the motivation and brief the methodology such that the contribution and the novelty of this work can be explained. It is not a literature review section, no need to put all the related works in it, just the ones can complete the logic and the story.

The logic is case by case, but it could follow the following way:

problem definition -> observation -> hypothesis -> methodology -> theory/development

Therefore, the paragraphs of the introduction could be arranged in this way (Note: each section should be smoothly connected, so that the logic is clear):

P1. Introduction of the big picture and the background of the topic focused in the paper (highlight the impact of the topic)

P2. Narrow down the introduction to the specific domain this paper in and conclude some existing limitations.

P3. Point out your observations (if any) on how the limitations could be tackled and give the hypothesis (if any) that should be tested for verifying the guesses. (focus on the scientific contribution of the work)

P4. Briefly explain the methodology of this paper, and then draw out the technical contributions.

P5. Outline of the paper

Hypothesis

Read this website to learn more about how to form a good hypothesis:
https://www.verywellmind.com/what-is-a-hypothesis-2795239

In short, if a hypothesis was not falsified after rigorous testing, it should ideally become a theorem or a proposition. Therefore, one way to check if a hypothesis is good is to read it as a theorem and see if it is meaningful. In addition, a hypothesis normally describes certain relationships, and thus the outcome of testing a hypothesis shouldn't be just a yes or no answer.

Literature review

This section reviews the most recent related works, which needs to serve two main purposes. First, it briefly details what have been done in the literature, so it is good to use one or two sentences to describe each work. Second, it brings out the knowledge gap and the importance of the present paper, so it is important to point out what is missing at the end of the section or each paragraph. A research paper normally has around 30 citations, and one may have two to three related areas to be reviewed.

Research methodology or approach

The subsections should be corresponded to the contributions.

Findings

If there are numerical testing, remember to put down the specification of the testing platforms. For example, a PC running 64-bit Windows 11 equipped with Intel Core i5-6500 CPU@3.20GHz, and 8GB RAM.

Conclusion

Summarize the paper and list some limitations.

If it is a material science (or mostly experimental) paper, it will typically include the following sections:

Introduction: a concise, up-to-date description of the background to provide a general reader of the Journal with enough context to understand the research being presented and its significance, as well as providing a clear statement of the research question and any hypotheses being explored.

Materials and methods: techniques, materials and equipment described in sufficient detail for another trained researcher to be able to reproduce the experimental work reported. Methods that are identical to published procedures should still be summarized in brief and include a citation to the original work.

Analysis: in submissions that have a significant theoretical or mathematical component, a description of the analytical procedures may be required.

Results: a description of the analyses and measurements related to answering the central research questions.

Discussion: the interpretation of the results, considering their significance and putting them into a wider context through comparison to previously published research. The use of a combined “Results and Discussion” section is discouraged.

Conclusions: a concise statement of the main conclusions drawn from the research reported in the manuscript.